

## Education

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| <b>Saintgits College of Engineering</b><br>B.Tech in Electronics and Computer Engineering | 2023 – Present<br>CGPA: 7.13 |
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## Skills

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| <b>Machine Learning / CV:</b> | PyTorch, YOLOv8 / YOLO11, CNNs, OpenCV, ONNX           |
| <b>Embedded / Edge:</b>       | ESP32, Raspberry Pi, Linux, Edge Computing, PlatformIO |
| <b>Programming:</b>           | Python, C/C++, SQL                                     |
| <b>Hardware &amp; Tools:</b>  | KiCad, Verilog, Cadence, Git, Streamlit                |

## Experience

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| <b>Embedded Systems Intern</b> — Keltron                                                                                                                                                                                    | 2 weeks, 2024 |
| <ul style="list-style-type: none"><li>Worked on microcontroller-based hardware interfacing and real-world embedded applications.</li><li>Gained exposure to industry workflows in electronics system development.</li></ul> |               |

## Projects

### HelmetGuard — Real-Time Helmet Detection

*YOLO11s · PyTorch · OpenCV · Streamlit*

- Trained a custom **YOLO11s** model on a Roboflow dataset across three classes (`helmet`, `without_helmet`, `two_wheeler`).
- Built an interactive **Streamlit** app supporting image, video, and live webcam inference modes.
- Optimised the real-time webcam inference pipeline for low latency using **PyTorch** and **OpenCV**.

### Smart Glasses — OCR-Based Assistive Reading

*Raspberry Pi Zero W · EasyOCR · pyttsx3 · Bluetooth Audio*

- Designed a wearable edge device that captures text via camera, runs on-device OCR, and outputs speech over Bluetooth.
- Deployed **EasyOCR** + **pyttsx3** fully offline on a **Raspberry Pi Zero W**, achieving real-time text-to-speech.
- Optimised the pipeline for low-power operation with minimal latency on constrained hardware.

### Tomato Leaf Disease Classifier — Handheld Edge Device

*YOLOv8s · MobileNetV3 · ONNX · Raspberry Pi 4B · TFT Display*

- Used **YOLOv8s** to detect and crop the diseased leaf region, feeding localised patches into a **MobileNetV3** classifier for accurate disease identification.
- Deployed the two-stage inference pipeline via **ONNX Runtime** on a **Raspberry Pi 4B+**.
- Integrated a TFT display and camera module to create a self-contained handheld diagnostic device.
- Achieved real-time inference on edge hardware without cloud dependency.

### RFID Attendance System — Distributed Edge IoT

*ESP32 · MQTT · Google Sheets API*

- Architected a multi-node **ESP32** RFID system with inter-device communication via a central edge coordinator.
- Automated attendance logging to **Google Sheets** in real time via API integration.
- Demonstrated a full IoT pipeline: sensor capture → edge processing → cloud sync.

### 32-bit Ripple Carry Adder

- Designed and simulated a 32-bit ripple carry adder in **Verilog**; verified via testbenches and waveform analysis.
- Analysed propagation delay behaviour to understand timing trade-offs in combinational logic.

## Certifications & Training

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- **Hands-on Introduction to Linux Commands and Shell Scripting** — IBM, Coursera *Oct 2025*
- **Hands-on Workshop on Vega Processor** — IIC & CDAC Trivandrum *May 2024*
- **Winter Technical Training Programme** — India Space Lab *Feb 2026*  
Drone Technology, CanSat/CubeSat, Rocketry, Remote Sensing (GIS)
- **Robotic Process Automation Workshop** — Automation Anywhere *Feb 2026*